

Can the claim survive the evidence?

Learning intention

Use evidence about control, circulation and gas exchange to judge claims in an unfamiliar model.

A fictional teaching lab has three connected models.

Your job: calculate, explain and limit its claims.

Recover three mechanisms

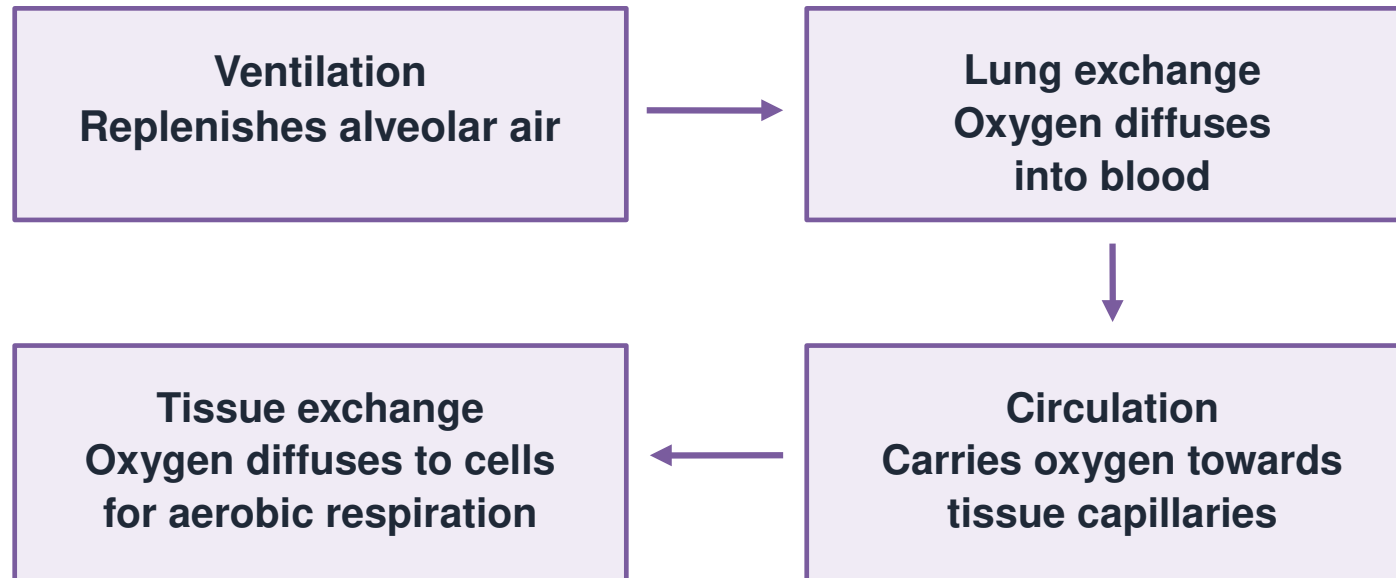
R1. What is cardiac output calculated from?

R2. Which way does net diffusion occur?

R3. Is insulin a signal or a glucose store?

Think across the oxygen pathway

Oxygen supply is a linked pathway



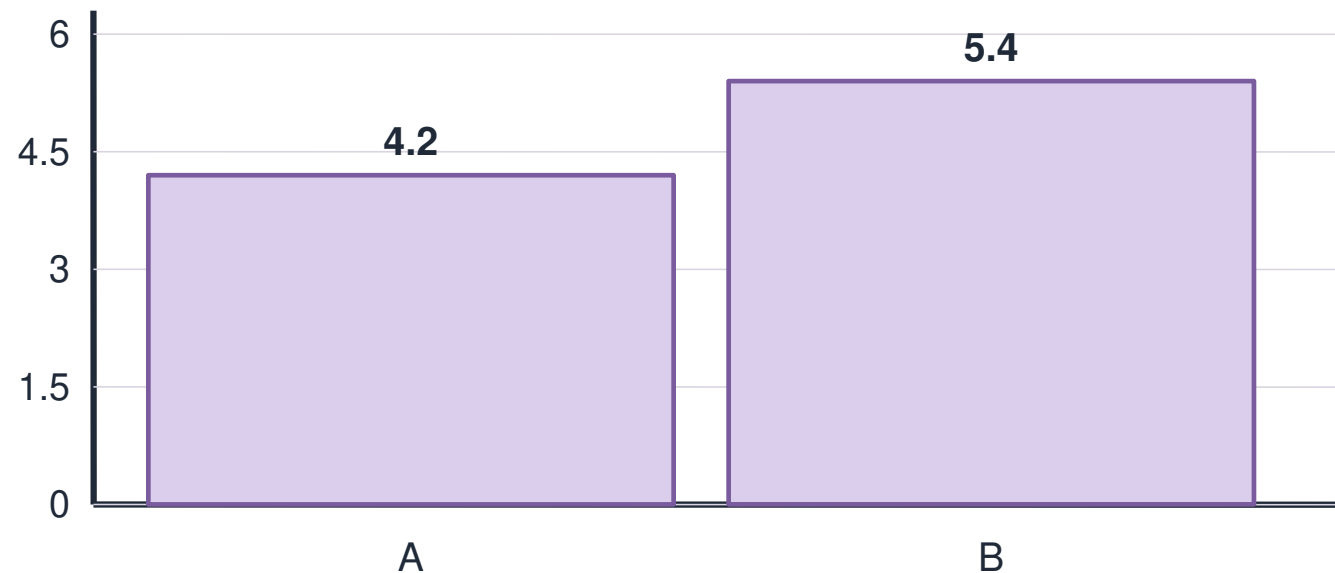
Each link has a different mechanism; one number cannot describe them all.

Name the job at each link. An output number describes one link only.

Calculate before judging

Constructed pump records: calculate both factors

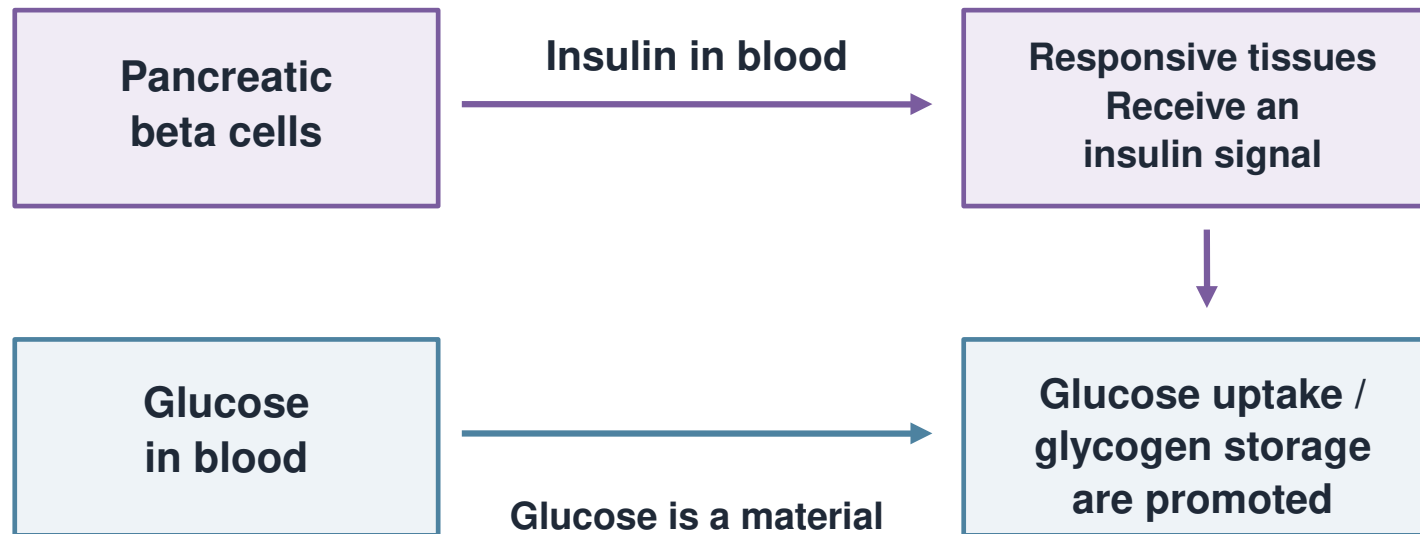
Model pump output (L/min)



A: $60 \times 70 = 4,200$ mL/min. Convert: $4,200 \div 1,000 = 4.2$ L/min.

Keep signal and material separate

Distinguish a hormone message from a transported fuel



Insulin is a signal; it is not converted into glycogen.

Insulin changes responses in target tissues. Glucose is transported and can be stored as glycogen.

Choose a fair comparison

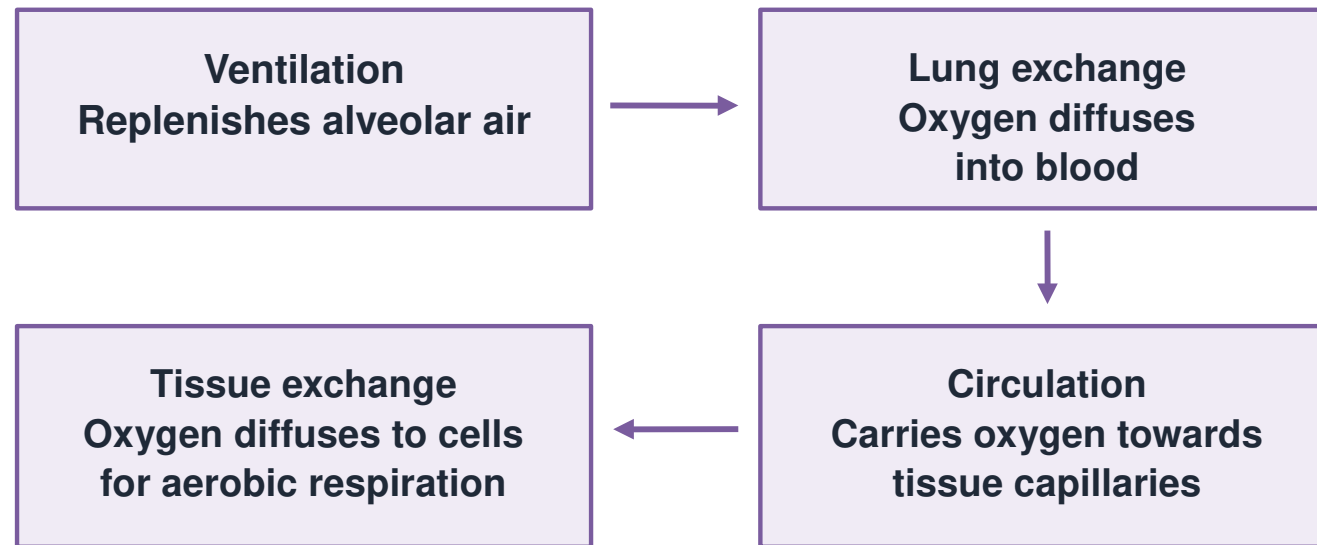
W1. Which pair tests area alone? W2. Which pair tests thickness alone?

Case	Relative area	Relative concentration difference	Relative thickness
X	3	4	1
Y	6	4	2
Z	6	4	1

Constructed model: rate index = area $\times$ concentration difference $\div$ thickness; same omitted constant.

Audit the headline

Oxygen supply is a linked pathway



Each link has a different mechanism; one number cannot describe them all.

W3. “B has a higher output, so every tissue definitely gets enough oxygen.” Keep the supported part; reject the unsupported part.

Test a tempting shortcut

“Y has twice the area of X, so its index must be twice as large.”

Predict, calculate, then explain.

Which conclusion is fully supported?

Select the claim that respects the evidence.

- A** Z has twice X's relative index because area doubles while concentration difference and thickness stay the same.
- B** B proves that a pupil is fitter than another pupil.
- C** Insulin is broken down into glycogen to provide a store.

Write a defensible evidence note

Choose one route.

Include a calculation, a mechanism and a limit.

Use the data; qualify the claim

Cite a named record or controlled pair.

Explain what the model cannot establish.

Audience: ask for the missing evidence

L1. Share one claim with a quiet partner or adult.

Audience asks: what would you need to measure next?

Make one claim stronger

Add a unit, a controlled comparison or a model limit.

Three quick evidence checks

- E1. Calculate $80 \text{ beats/min} \times 50 \text{ mL/beat}$ in L/min.
- E2. Why is X/Y not an area-only test?
- E3. Name one missing quantity for oxygen adequacy.

Choose the next practice

Units • controlled pairs • hormone roles • limits.